P3

Seri 7 Pertemuan

PROYEK GAME 2D · PERTEMUAN 3

# Animasi, Kamera & Interaksi

Politeknik Negeri Batam — Program Studi Teknologi Permainan

MELANJUTKAN P2

# Yang sudah ada, dan yang kita tambahkan hari ini

### Sudah ada dari P2

**PlayerMovement** v4 + **IMoveInput** + **Cfg_Player**

Scn_Bootstrap → Scn_World, **WorldInstaller**

Peta 40 × 36: Tilemap, Rule Tile, sorting, Pixel Perfect

### Ditambahkan hari ini

8 animation clip + Blend Tree

PlayerMovement v5 + PlayerAnimator

Kamera Cinemachine yang dibatasi peta

Tombol Interact, NPC prefab & varian, AreaZone

TUJUAN PEMBELAJARAN

# Setelah pertemuan ini, mahasiswa mampu...

| # | Kemampuan | Bukti di akhir kelas |
|---|---|---|
| 1 | Membuat animation clip dari sprite sheet | 8 clip Anim_Player_* dengan Samples 8 |
| 2 | Menyusun Animator dengan Blend Tree 2D | Idle ↔ Walk instan, arah dari MoveX/MoveY |
| 3 | Memisahkan logika gerak dan tampilan | PlayerAnimator membaca PlayerMovement |
| 4 | Memasang Cinemachine untuk pixel art | Kamera mengikuti, berhenti di tepi peta |
| 5 | Membuat interaksi lewat interface | Tekan E: NPC menoleh dan menyapa |
| 6 | Membuat varian prefab | 4 warga dari satu Pfb_Npc |

AGENDA · ±150 MENIT

# Enam praktikum, dari sprite diam sampai dunia yang menyapa

| Praktikum | Durasi | Hasil |
|---|---|---|
| 1 · Animation clip | 20 menit | 8 clip diam & jalan × 4 arah |
| 2 · Animator & Blend Tree | 25 menit | AC_Player: Idle ↔ Walk |
| 3 · Kode animasi | 20 menit | PlayerMovement v5 + PlayerAnimator |
| 4 · Cinemachine + Confiner | 30 menit | CM_Follow, CameraBounds |
| 5 · Layer & interaksi | 30 menit | IActionInput, IInteractable, PlayerInteractor |
| 6 · NPC prefab & AreaZone | 20 menit | Tetua, Sari, Penjaga, Pak Dagang; 4 area |
| Uji & kirim | 5 menit | Checklist + PR |

GAMBARAN AKHIR

# Kamera berhenti di tepi peta, Player tetap beranimasi




Player berjalan ke kanan: sprite **_8**, state **Walk**.

Kamera berhenti di **x = 30** (lebar peta 40 - setengah layar 10).

Tidak ada latar kosong di luar peta.

PRAKTIKUM 1 · 20 MENIT

# Animation Clip

PRAKTIKUM 1 · PETA FRAME

# Clip = sprite yang berganti menurut waktu

| Arah | Anim_Player_Idle_... | Anim_Player_Walk_... (Samples 8) |
|---|---|---|
| Down | _0 | _0, _1, _0, _2 |
| Left | _3 | _3, _4, _3, _5 |
| Right | _6 | _6, _7, _6, _8 |
| Up | _9 | _9, _10, _9, _11 |

**1.** Pilih Player → Window → Animation → **Animation** → Create

**2.** Clip pertama otomatis membuat Animator → ganti nama **AC_Player**

**3.** Walk: Samples **8**, Loop Time aktif



PRAKTIKUM 2 · 25 MENIT

# Animator & Blend Tree

PRAKTIKUM 2 · LANGKAH

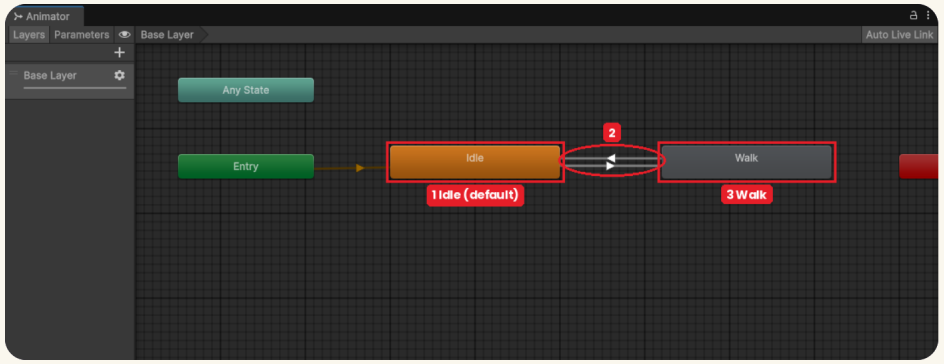
# AC_Player: dua state, masing-masing berisi 4 arah

**1.** Parameter: **MoveX**, **MoveY** (Float), **IsMoving** (Bool)

**2.** Create State → From New Blend Tree: **Idle** dan **Walk**

**3.** Blend Tree: **2D Simple Directional**, motion di (0,-1) (-1,0) (1,0) (0,1)

**4.** Transisi IsMoving true/false, **Has Exit Time mati**, **Duration 0**



PRAKTIKUM 3 · 20 MENIT

# Kode Animasi

PRAKTIKUM 3 · LANGKAH 1 · VERSI 5

# PlayerMovement.cs (bagian yang berubah)

```
public Vector2 Velocity => velocity;
public Vector2 Facing { get; private set; } = Vector2.down;

void Update()
{
    if (input == null) return;
    Vector2 move = input.Move;
    float speed = input.IsRunning ? config.RunSpeed :
            config.WalkSpeed;
    velocity = move * speed;
    if (move.sqrMagnitude > 0.01f) Facing = ToCardinal(move);
}

private static Vector2 ToCardinal(Vector2 v) =>
    Mathf.Abs(v.x) > Mathf.Abs(v.y)
        ? new Vector2(Mathf.Sign(v.x), 0f)
        : new Vector2(0f, Mathf.Sign(v.y));
```

| Input | Facing |
|---|---|
| (1, 0.2) | (1, 0) kanan |
| (-0.3, -0.9) | (0, -1) bawah |
| (0.7, 0.7) | (0, 1) atas |

**Facing** tetap menyimpan arah terakhir saat berhenti

PRAKTIKUM 3 · LANGKAH 2 · SCRIPTS/PLAYER

# PlayerAnimator.cs

```
[RequireComponent(typeof(Animator))]
public class PlayerAnimator : MonoBehaviour
{
    private static readonly int MoveX = Animator.StringToHash("MoveX");
    private static readonly int MoveY = Animator.StringToHash("MoveY");
    private static readonly int IsMoving = Animator.StringToHash("IsMoving");

    [SerializeField] private PlayerMovement movement;
    private Animator animator;

    void Awake() => animator = GetComponent<Animator>();

    void Update()
    {
        animator.SetBool(IsMoving, movement.Velocity.sqrMagnitude > 0.01f);
        animator.SetFloat(MoveX, movement.Facing.x);
        animator.SetFloat(MoveY, movement.Facing.y);
    }
}
```

**StringToHash**: nama parameter diubah ke angka sekali saja

Satu script satu tugas: animator hanya **membaca**

Hasil uji: berjalan ke kanan → state Walk, sprite _8, MoveX 1

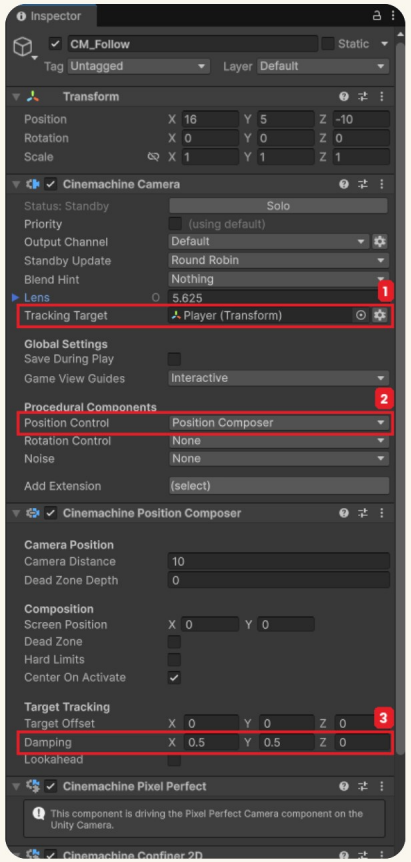
PRAKTIKUM 4 · 30 MENIT

# Cinemachine

PRAKTIKUM 4 · LANGKAH

# CM_Follow tanpa satu baris kode

1. Package Manager → **Cinemachine** (6000.6: versi 6.6.0)
2. Main Camera: Add Component → **Cinemachine Brain**
3. GameObject kosong **CM_Follow** → **Cinemachine Camera**, Tracking Target **Player**
4. Lens → **Orthographic Size 5,625** (bawaan 10 merusak Confiner)
5. Position Control **Position Composer** (Distance 10, Damping 0.5/0.5/0), Rotation **None**
6. Add Extension → **Cinemachine Pixel Perfect**
7. Add Extension → **Confiner 2D**, Bounding Shape 2D = **CameraBounds**



PRAKTIKUM 4 · CAMERABOUNDS

# Pagar tak terlihat untuk kamera

**1.** GameObject kosong **CameraBounds** di (0, 0, 0), layer **CameraBounds**

**2. Polygon Collider 2D**, ubah menjadi 4 titik: (0,0) (40,0) (40,36) (0,36)

**3.** Centang **Is Trigger** agar tidak menghalangi Player

**4.** Mengubah titik? tekan **Invalidate Bounding Shape Cache**

### Hitungan jendela kamera

Layar 320 × 180 piksel ÷ PPU 16 = **20 × 11,25 unit**.

Hasil uji: kamera berhenti di x = 30, y = 5,63 saat Player di sudut kanan bawah.

PRAKTIKUM 5 · 30 MENIT

# Interaksi

PRAKTIKUM 5 · LANGKAH 1

# Layer untuk seluruh semester

| Layer | Dipakai oleh | Collision Matrix |
|---|---|---|
| Player | Player | normal |
| Interactable | NPC, Tetua, Batu Simpan | normal |
| Wild | Monster liar (P4) | normal |
| CameraBounds | Batas kamera | tidak bertabrakan dengan apa pun |

PRAKTIKUM 5 · LANGKAH 2

# IActionInput, IInteractable, PlayerInteractor

```
public interface IActionInput { bool InteractPressed { get; } }
public interface IInteractable { void Interact(PlayerInteractor who); }

public class PlayerInteractor : MonoBehaviour
{
    [SerializeField] private PlayerMovement movement;
    [SerializeField] private float reach = 0.6f;
    [SerializeField] private float radius = 0.3f;
    [SerializeField] private LayerMask interactableMask;
    private IActionInput input;

    public void Construct(IActionInput actionInput) => input = actionInput;
    private Vector2 ProbePoint => (Vector2)transform.position + movement.Facing * reach;
```

PRAKTIKUM 5 · LANJUTAN

# IActionInput, IInteractable, PlayerInteractor (lanjutan)

```
    void Update()
    {
        if (input == null || !input.InteractPressed) return;
        Collider2D hit = Physics2D.OverlapCircle(ProbePoint, radius, interactableMask);
        if (hit != null && hit.TryGetComponent(out IInteractable target))
            target.Interact(this);
    }
}
```

PRAKTIKUM 5 · LANGKAH 2B · SCRIPTS/CORE/INPUT

# InputSystemActionInput.cs

```
using UnityEngine.InputSystem;

public class InputSystemActionInput : IActionInput
{
    private readonly InputAction interact;

    public InputSystemActionInput()
    {
        interact = InputSystem.actions.FindAction("Interact");
    }

    public bool InteractPressed ⇒ interact.WasPressedThisFrame();
}
```

Pasangan IMoveInput dari P2: class C# biasa

**WasPressedThisFrame**: true tepat satu frame saat E ditekan

Action Interact bawaan memakai interaction **Hold**: biarkan, cara ini tetap bereaksi saat tombol baru ditekan

PRAKTIKUM 5 · LANGKAH 3 · SCRIPTS/CORE/BOOTSTRAP

# Composition Root membuat service kedua

```
public class GameBootstrap : MonoBehaviour
{
    [SerializeField] private string firstScene = "Scn_World";
    private IMoveInput moveInput;
    private IActionInput actionInput;
    void Awake()
    {
        moveInput = new InputSystemMoveInput();
        actionInput = new InputSystemActionInput();
    }
    IEnumerator Start()
    {
        yield return SceneManager.LoadSceneAsync(firstScene, LoadSceneMode.Additive);
        SceneManager.SetActiveScene(SceneManager.GetSceneByName(firstScene));
        FindAnyObjectByType<WorldInstaller>().Install(moveInput, actionInput);
    }
}
```

PRAKTIKUM 5 · LANGKAH 3 · SCRIPTS/WORLD

# WorldInstaller menerima service kedua

```
public class WorldInstaller : MonoBehaviour
{
    [SerializeField] private PlayerMovement player;
    [SerializeField] private PlayerInteractor interactor;

    public void Install(IMoveInput moveInput, IActionInput actionInput)
    {
        player.Construct(moveInput);
        interactor.Construct(actionInput);
    }
}
```

PRAKTIKUM 6 · 20 MENIT

# Prefab & Area

PRAKTIKUM 6 · LANGKAH 1 · SCRIPTS/WORLD

# Npc.cs versi 1

```
public class Npc : MonoBehaviour, IInteractable
{
    [SerializeField] private string npcName = "Warga";
    [SerializeField] private SpriteRenderer body;
    [SerializeField] private Sprite faceDown, faceLeft, faceRight,
            faceUp;

    public void Interact(PlayerInteractor who)
    {
        Face(who.transform.position - transform.position);
        Debug.Log($"{npcName}: Halo!");
    }

    private void Face(Vector2 d) ⇒
        body.sprite = Mathf.Abs(d.x) > Mathf.Abs(d.y)
            ? (d.x > 0 ? faceRight : faceLeft)
            : (d.y > 0 ? faceUp : faceDown);
}
```

**: MonoBehaviour, IInteractable**: satu induk, boleh banyak interface

Di P6 Debug.Log diganti dialog sungguhan

Versi demo (P6): Interact memanggil **Dialogue.Run(Talk())**, dan **Talk() virtual** ditulis ulang oleh turunan

PRAKTIKUM 6 · LANGKAH 1B · SCRIPTS/WORLD

# AreaZone.cs versi 1

```
using UnityEngine;

[RequireComponent(typeof(Collider2D))]
public class AreaZone : MonoBehaviour
{
    [SerializeField] private string areaName = "Desa Awal";
    public string AreaName ⇒ areaName;

    void OnTriggerEnter2D(Collider2D other)
    {
        if (!other.CompareTag("Player")) return;
        Debug.Log($"Masuk area: {areaName}");
    }
}
```

Pasang di GameObject dengan **Box Collider 2D Is Trigger**

Player wajib ber-tag **Player**

P5 menambah event Entered (HUD), P6 menambah musik area

PRAKTIKUM 6 · LANGKAH 2

# Satu prefab, empat varian

**1.** Rakit NPC: layer **Interactable**, Sorting Layer **Characters**, Box Collider 2D (0.8 × 0.5) di kaki

**2.** Seret ke **Prefabs/NPC** → **Pfb_Npc**

**3.** Klik kanan → Create → **Prefab Variant**: Elder, Girl, Guard, Merchant

**4.** Isi nama + sprite hadap _0/_3/_6/_9; letakkan sesuai peta (slide berikut)

**5.** 4 **AreaZone** (Box Collider 2D trigger): Desa, Hutan, Danau, Gua



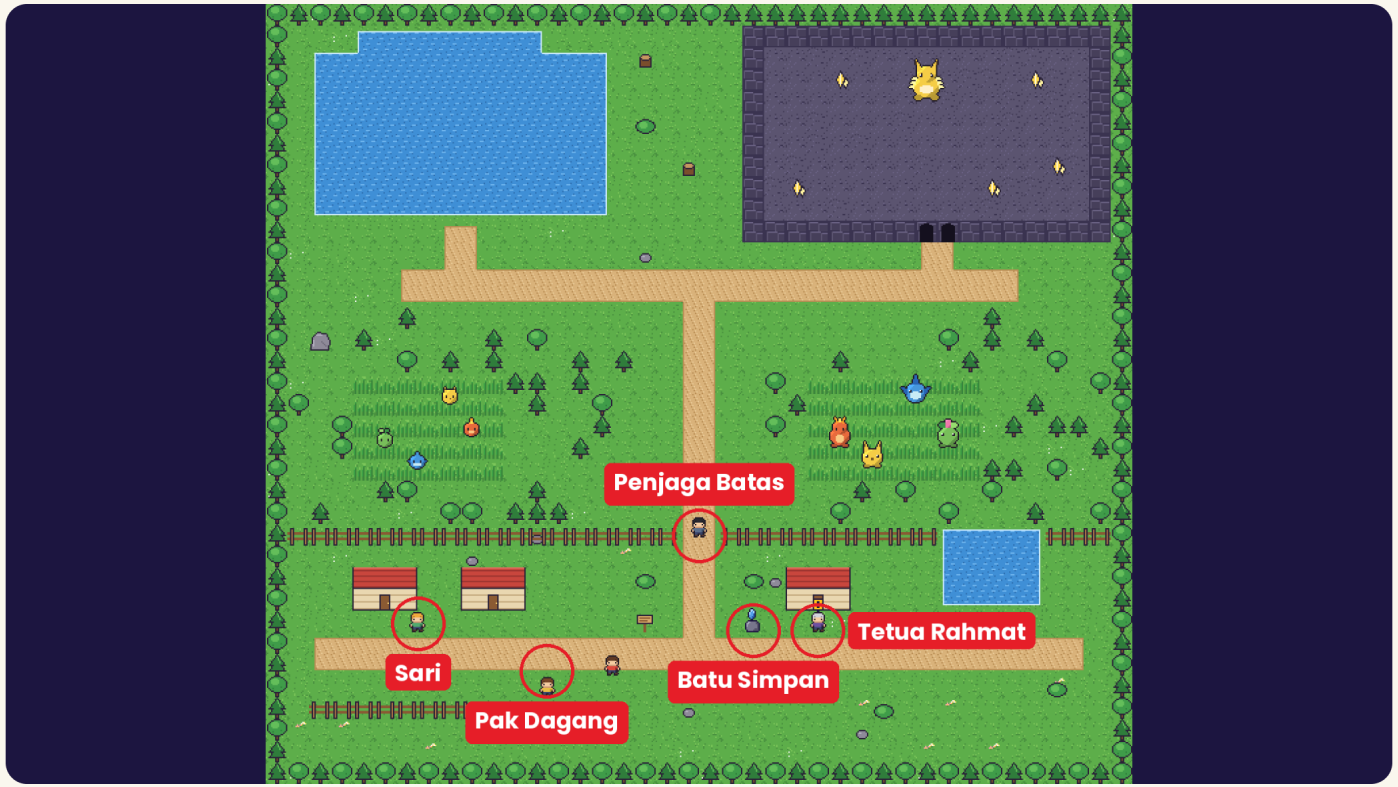
PRAKTIKUM 6 · LANGKAH 3 · REVISI DEMO

# Empat warga desa

| Objek (varian) | npcName | Posisi | Peran nanti |
|---|---|---|---|
| Pfb_Npc_Elder | Tetua Rahmat | (25,5; 7) depan rumah kanan | memberi partner (P6: StarterGiver) |
| Pfb_Npc_Girl | Sari | (7; 7) depan rumah kiri | tips elemen (DialogueData) |
| Pfb_Npc_Guard | Penjaga Batas | (20; 11,4) celah pagar | menahan jalan ke hutan (P6: GuardNpc) |
| Pfb_Npc_Merchant | Pak Dagang | (13; 4) jalan desa bawah | hadiah 3 Ramuan (P6: MerchantNpc) |

PRAKTIKUM 6 · HASIL

# Peta NPC versi demo




**Penjaga Batas** berdiri di celah pagar

**Tetua Rahmat** di kanan jalan desa

**Sari** dan **Pak Dagang** di kiri

Batu Simpan (P6) di samping Tetua

PRAKTIKUM 6 · KE DEPAN

# Npc yang sama di P6 dan demo




Tetua menoleh lalu berbicara




Pak Dagang memberi 3 Ramuan

UJI DARI SCN_BOOTSTRAP

# Checklist lulus pertemuan 3

### Checklist lulus

- Animasi sesuai arah, diam = pose arah terakhir
- Kamera halus, tepi peta tidak kosong
- E di depan NPC → NPC menoleh + log
- 4 NPC bernama sesuai demo
- Masuk area tercatat
- Console bersih

| Gejala | Perbaikan |
|---|---|
| Selalu menghadap bawah | Nama parameter Animator ≠ kode |
| Animasi telat berganti | Has Exit Time mati, Duration 0 |
| Kamera diam | Tracking Target kosong / Brain tidak ada |
| Kamera keluar peta | Bounding Shape kosong / cache lama |
| Kamera terkunci di tengah | Lens 5,625 lalu Invalidate Bounding Shape Cache |
| E tidak bereaksi | NPC bukan layer Interactable / mask |

TUGAS INDIVIDU · TENGGAT [TANGGAL]

# Animasi untuk karakter lain

**1.** Buat **AOC_Npc_Elder** (Animator Override Controller dari AC_Player) dengan clip NPC_Elder

**2.** Clip idle 2 frame untuk 4 monster tahap 1 (Samples 3) — dipakai di P4

**3.** Tantangan: animasi jalan 1,5× lebih cepat saat lari (parameter Speed + Multiplier)

```
art: animation clip & AC_Player
```

```
feat: PlayerAnimator & movement v5
```

```
feat: kamera Cinemachine + confiner
```

```
feat: IInteractable & prefab NPC
```

P4

BERIKUTNYA · P4

# Data Monster & Monster Liar

- 12 MonsterData dari CSV + MonsterDatabase
- MonsterInstance: level, EXP, evolusi
- FSM monster liar & transisi ke battle